

Further information needed:

Minimum cardinality assumptions:

1. An EMPLOYEE can exist in the system without having been assigned to a JOB. For example, a manager or a new employee in the company. Hence, an EMPLOYEE can be associated with zero or many JOBS.
2. An EMPLOYEE at the top of the hierarchy (e.g., the owner) is not managed by anyone. Hence, the unary relationship on EMPLOYEE must allow for an employee to be managed by zero or one manager.
3. A COMBINATION of a truck and trailer can be defined before it is ever assigned to a JOB. Hence, a COMBINATION can be associated with zero or many JOBS.
4. An EMPLOYEE can exist in the system without having prepared any QUOTES. For example, a new employee or a mechanic who doesn't prepare quotes. Hence, an EMPLOYEE can be associated with zero or many QUOTES.

Surrogate key justification:

The surrogate key is combination_id. This surrogate key replaces the composite natural key, which is a combination of purpose_no and combination_code, that uniquely identifies a specific hire rate for a combination used for a particular purpose. A single, numeric primary key (combination_id) is simpler and more efficient for other tables, like JOB, to reference as a foreign key compared to a multi-column composite key.